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Higher ratio of plasma omega-6/omega-3 fatty acids is associated with greater risk of all-cause, cancer, and cardiovascular mortality: a population-based cohort study in UK Biobank

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Abstract

Background

Circulating omega-3 and omega-6 polyunsaturated fatty acids (PUFAs) have been associated with various chronic diseases and mortality, but results are conflicting. Few studies examined the role of a balanced omega-6/omega-3 ratio in mortality.

Methods

We investigated plasma omega-3 and omega-6 PUFAs and their ratio in relation to all-cause and cause-specific mortality in a large prospective cohort, the UK Biobank. Of 85,425 participants who had complete information on circulating PUFAs, 6,461 died during follow-up, including 2,794 from cancer and 1,668 from cardiovascular disease (CVD). Associations were estimated by multivariable Cox proportional hazards regression with adjustment for relevant risk factors.

Results

Risk for all three mortality outcomes increased as the ratio of omega-6/omega-3 PUFAs increased (all $P_{\text{trend}} < 0.05$). Comparing the highest to the lowest quintiles, individuals had 26% (95% CI, 15-38%) higher total mortality, 14% (95% CI, 0-31%) higher cancer mortality, and

31% (95% CI, 10-55%) higher CVD mortality. Moreover, omega-3 and omega-6 PUFAs in plasma were all inversely associated with all-cause, cancer, and CVD mortality, with omega-3 showing stronger effects.

Conclusions

Using a population-based cohort in UK Biobank, our study revealed a strong association between the ratio of circulating omega-6/omega-3 PUFAs and the risk of all-cause, cancer, and CVD mortality.

eLife assessment

This **important**, well-conducted study in a large data set - the UK BioBank population - reports that both circulating omega-6 and omega-3 PUFAs as well as the ratio of omega-6 to omega-3 PUFAs are associated with lower all-cause, cancer and cardiovascular mortality. The study is **convincing** and these findings will be of broad interest to epidemiologists, nutritionists, medical practitioners and the general population.

Background

Cancer and cardiovascular disease (CVD) are the two leading causes of non-communicable disease mortality globally.¹ Substantial epidemiologic evidence has linked the dietary or circulating levels of omega-3 and omega-6 polyunsaturated fatty acids (PUFAs) to the risks of all-cause, cancer, and CVD mortality (**Additional file 1: Table S1**). However, the results are inconsistent, especially for omega-6 PUFAs. Most large observational studies support the protective effects of omega-3 PUFAs. In addition to total and individual omega-3 PUFAs, the omega-3 index, defined as the percentage of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) in total fatty acids in red blood cells, was shown to be a validated biomarker of the dietary intake and tissue levels of long-chain omega-3 PUFAs, and was proposed to be a risk factor for CVD and related mortality.^{2,3} While the omega-3 index has been observed to be inversely associated with all cause-mortality, its association patterns with CVD and cancer mortality are less clear.⁴⁻⁶ Most importantly, results from clinical trials of omega-3 PUFA supplementation have been inconsistent.^{7,8} On the other hand, while some studies revealed inverse associations of omega-6 PUFAs with all-cause mortality,⁹⁻¹¹ others reported null results.^{4,12-15} The role of omega-6 PUFAs in cancer and CVD mortality is less studied, and the patterns are similarly conflicting.^{4,9-11} Interpreting previous observational results is challenging due to the limitations of small sample sizes, insufficient adjustments for confounding, and unique sample characteristics. Moreover, many studies rely on self-reported dietary intake or fish oil supplementation status, which are subject to large variability and reporting bias.¹⁶ Despite the tremendous interest and research effort, the roles of omega-3 and omega-6 PUFAs in all-cause and cause-specific mortality remain controversial.

It has been suggested that the high omega-6/omega-3 ratio in Western diets, 20:1 or even higher, as compared to an estimated 1:1 during the most time of human evolution, contributes to many chronic diseases, including CVD, cancer, and autoimmune disorders.^{17,18} However, while many previous studies have examined total or individual omega-3 and omega-6 PUFAs, fewer investigated the role of their imbalance, as measured by the omega-6/omega-3 ratio, in

mortality^{4,14,15,19}. In a prospective cohort study of postmenopausal women, the omega-6/omega-3 ratio in red blood cells was associated with an increased risk of all-cause mortality but not with cancer or CVD mortality⁴. Similar positive associations with all-cause mortality were observed in smaller cohorts investigating the ratio in serum or dietary intake^{14,19}. However, prospective studies in two independent cohorts from China and US did not find a linear positive association between the omega-6/omega-3 ratio in diet and all-cause mortality¹⁵. To address these gaps in our understanding of the roles of omega-3 and omega-6 PUFAs and their imbalance in all-cause and cause-specific mortality, we perform a prospective study in a large population-based cohort (N = 85,425) from UK Biobank, using objective measurements of PUFA levels in plasma.

Methods

Study population

The UK Biobank study is a prospective, population-based cohort study in the United Kingdom²⁰. Between 2006 and 2010, 502,384 prospective participants, aged 40-69, in 22 assessment Centers throughout the UK were recruited for the study. The population information was collected through a self-completed touch-screen questionnaire; brief computer-assistant interview; physical and functional measures; and blood, urine, and saliva collection during the assessment visit. Participants with cancer (n=37,736) or CVD (n=100,972), those who withdrew from the study (n=879), and those with incomplete data on the plasma omega-6/omega-3 ratio (n=277,372) were excluded from this study, leaving 85,425 participants, 6,461 died during follow-up, including 2,794 from cancer and 1,668 from CVD.

Ascertainment of exposure

Plasma PUFA levels (absolute concentrations and percentages of total fatty acids) were measured by nuclear magnetic resonance (NMR) in plasma samples collected between 2007 and 2010, including total PUFAs, omega-3 PUFAs, omega-6 PUFAs, docosahexaenoic acid (DHA), and linoleic acid (LA)²⁰⁻²².

Ascertainment of outcome

The date and cause of death were identified through the death registries of the National Health Service (NHS) Information Centre for participants from England and Wales and the NHS Centre Register Scotland for participants from Scotland²⁰. At the time of the analysis (15 Mar 2023), we had access to the most up-to-date mortality dataset (Version: December 2022), which had the date of death up to 12 November 2021. Therefore, follow-up time was calculated as the time between the date of entering the assessment centre and this date, or the date of death, whichever happened first. The underlying cause of death was assigned and coded in vital registries according to the International Classification of Diseases, 10th revision (ICD-10). CVD mortality was defined using codes I00-I99, and cancer mortality was defined using codes C00-D48. To exclude those who had cancer at baseline, we used the overall cancer incidence information based on diagnoses in cancer registers derived from the Health and Social Care Information Centre and the NHS Central Register that were coded based on ICD-9 and ICD-10 codes. The information about CVD incidence was retrieved from Hospital inpatient data and Death Register records that were coded according to ICD-9 and ICD-10 codes, and self-reported medical condition.

Ascertainment of covariates

The baseline questionnaire included detailed information on several possible confounding variables: demographic factors (age, sex, assessment centre, ethnicity), socioeconomic status (Townsend Deprivation Index), lifestyle habits (alcohol assumption, smoking status, body mass

index (BMI), physical activity, comorbidities (including hypertension, diabetes, and longstanding illness)), and other supplementation (Fish-oil supplementation). The Townsend Deprivation Index, used as an indicator of socioeconomic status, is retrieved directly from the UK Biobank. BMI was defined as the body mass divided by the square of the body height and was expressed in units of kg/m^2 .²² Prevalent hypertension was derived from a self-reported history of hypertension. Information on diabetes and longstanding illness was collected by self-report at baseline. We also retrieved the information on dietary PUFA intakes and serum biomarkers for secondary analysis. Dietary PUFAs were estimated based on the 24-hour dietary recall.²³ The percentage of dietary omega-3 and omega-6 PUFA were calculated as the corresponding absolute dietary PUFA divided by summation of dietary monounsaturated fatty acids, saturated fatty acids, omega-3 fatty acids, and omega-6 fatty acids. The biochemistry markers were measured in the blood sample collected at recruitment.

Statistical analysis

We summarized and compared the characteristics of the participants across quintiles of the omega-6/omega-3 ratio at baseline using descriptive statistics. Pearson's Chi-squared test and ANOVA test were used to compare the demographic characteristics across quintiles, respectively, for categorical variables and continuous variables. To investigate associations of the ratio with all-cause and cause-specific mortality, we used multivariable Cox proportional hazards regression models to calculate hazard ratios²⁴ and their 95% confidence intervals (CI). The proportional hazards assumption was not violated based on Schoenfeld residuals. We analyzed the ratio as continuous and categorical variables (i.e., quintiles). For all trend tests, we used the median value of each quintile as a continuous variable in the models. Potential nonlinear associations were assessed semi-parametrically using restricted cubic splines (4 knots were used in regression splines).²⁵

Based on previous literature and biological plausibility^{11, 26, 27}, we chose the following variables as covariates in the multivariate models: age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m^2 ; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

In secondary analyses to assess potential differences in associations across different population subgroups, we repeated the above-described analyses stratified by age ($<$ vs. $\geq$ the median age of 58 years), sex (male/female), Townsend deprivation index ($<$ vs. $\geq$ the population median of -2), BMI ($<$ vs. ≥ 25), comorbidities (yes vs. no), physical activity (low and moderate vs. high) and current smoking status (yes vs. no). Besides, to compare the effects of dietary intake of PUFAs with the plasma levels on mortality, we repeated the above-described analyses in participants with complete information on dietary intake PUFAs ($N=153,064$). Moreover, we conducted mediation analysis to investigate potential biological pathways that might explain the effects of omega-6/omega-3 ratio on mortality outcomes. Based on plausibility, we chose Cholesterol, LDL, HDL-C, TG, ApoA, ApoB, CRP and Lp(a) as potential mediating factors for CVD mortality; SHBG, TTST, E2 and IGF-1 as potential mediating factors for cancer mortality; and all the above biomarkers for all-cause mortality. We first examined the variations in the concentrations of each biomarker according to omega-6/omega-3 ratio quintiles. Then we constructed two regression models, a linear model, to regress the mediator (treated the biomarkers as continuous) on the exposure (treated the omega-6/omega-3 ratio as continuous), with fully-adjustment for potential confounders; and a Cox proportional hazard model, to regress the outcome (mortality outcomes) on the exposure and the mediator, with fully-adjustment for potential confounders. We chose to construct Cox models for mediation analysis because our outcomes were rare (all $<10\%$).²⁸ We integrated these two regression models to obtain the indirect effects and proportions of mediating effects through the counterfactual-based approach proposed by Vander Weele.^{29, 30}

We also conducted several sensitivity analyses. First, to assess whether the associations of the omega-6/omega-3 ratio with mortality outcomes were primarily driven by omega-3 fatty acids or omega-6 fatty acids, we assessed both the separate and the joint associations of omega-3 fatty acids to total fatty acids percentage (omega-3%) and omega-6 fatty acids to total fatty acids percentage (omega-6%) with the three mortality outcomes. We also performed a joint analysis with categories of the omega-3% and omega-6% quintiles, using participants in both the lowest omega-3% and omega-6% quintiles as the reference category. An interaction term between omega-3% and omega-6% was included in the multivariable Cox proportional hazards model, and a Likelihood Ratio test was used to assess its significance. The correlation between omega-3% and omega-6% was assessed by the Pearson correlation. Second, to assess the effects of individual PUFAs on mortality, we conducted the same analysis for circulating docosahexaenoic acid to total fatty acids percentage (DHA%) and linoleic acid to total fatty acids percentage (LA%) with the three mortality outcomes. Third, to address the potential residual confounding by fish oil supplementation, we further adjusted for the supplementation status from the baseline questionnaire. Fourth, to investigate the effects of missing values, we imputed missing values (<1% for most factors, up to 19% for physical activity) by chained equations and performed synthesis analyses on the imputed datasets³¹. Fifth, to assess whether the observed associations are attenuated by reverse causation, we excluded those who died in the first year of follow-up. Last, to assess the representativeness of the participants included in our study, we compared the baseline characteristics between the participants with and without exposure information. All P-values were two-sided. We considered a P-value <0.05 or a 95% confidence interval (CI) excluding 1.0 for HRs as statistically significant. We conducted all analyses using R, version 4.0.3 and SAS statistical software, version 9.4 (SAS Institute Inc.).

Results

Baseline characteristics

In the analytic cohort of 85,425 participants, over a mean of 12.7 years of follow-up, 6,461 died, including 2,794 from cancer and 1,668 from CVD. The baseline characteristics of the participants across quintiles of the ratio of omega-6/omega-3 were summarized in **Table 1**. Study participants were, on average, 56 years old and 90% White. Those in the higher ratio quintiles were more likely to be younger, male, and current smokers, but less likely to have comorbidities and take fish oil supplementation.

Main results

The associations of the omega-6/omega-3 ratio with all-cause and cause-specific mortality risks were presented in **Table 2**. A higher ratio was strongly associated with higher mortality from all causes, cancer, and CVD ($P_{\text{trend}} < 0.05$ for all three). In the fully adjusted models that considered the ratio as a continuous variable, every unit increase in the ratio corresponded to 2%, 1%, and 2% higher risk in all-cause, cancer, and CVD mortality, respectively. When comparisons were made between the highest and the lowest quintile of the omega-6/omega-3 ratio, there were 26%, 14%, and 31% increased risk for all-cause, cancer, and CVD mortality, respectively.

Stratified analysis

The fully adjusted associations of the omega-6/omega-3 ratio with all-cause mortality revealed that compared to the lowest quintile, the highest quintile has strong, statistically significant associations with elevated risk within all categories of age, sex, TDI, BMI, comorbidities, physical activity, and smoking status (**Figure 1**, **Additional file 2: Table S2**), except in those aged less than 58 years old and BMI less than 25. The estimated associations with all-cause mortality were stronger in current smokers (P for interaction < 0.01; **Figure 1**, **Additional file 2: Table S2**). For

Characteristics ^a	Omega-6/omega-3 ratio quintiles					P
	1 (median = 5.9) (n = 17,085)	2 (median = 7.6) (n = 17,085)	3 (median = 9.1) (n = 17,085)	4 (median = 11.0) (n = 17,085)	5 (median = 14.8) (n = 17,085)	
Age (years)	58.6 (7.5)	57.2 (7.9)	55.7 (8.2)	54.7 (8.3)	53.4 (8.2)	<0.001 ^b
Sex (male%)	40.3	44.9	47.5	49.4	52.8	<0.001 ^c
Ethnicity (n%)						
White	15,375 (90.4%)	15,486 (91.0%)	15,494 (91.1%)	15,467 (91.1%)	15,420 (90.7%)	0.384 ^c
Black	113 (0.7%)	99 (0.6%)	105 (0.6%)	92 (0.5%)	116 (0.7%)	
Asian	673 (4.0%)	639 (3.8%)	663 (3.9%)	653 (3.8%)	659 (3.9%)	
Others	844 (5.0%)	786 (4.6%)	742 (4.4%)	773 (4.6%)	810 (4.8%)	
Missing (n)	86	75	81	106	86	
BMI	27.0 (4.4)	27.3 (4.5)	27.3 (4.7)	27.3 (4.8)	26.9 (5.0)	<0.001 ^b
Missing (n)	54	64	55	58	71	
TDI	-1.6 (3.0)	-1.5 (3.0)	-1.4 (3.1)	-1.2 (3.1)	-0.9 (3.2)	<0.001 ^b
Missing (n)	17	24	24	28	22	
Smoking status (n%)						<0.001 ^c
Never	9,426 (55.5%)	9,272 (54.6%)	9,424 (55.4%)	9,400 (55.3%)	9,214 (54.2%)	
Previous	6,402 (37.7%)	6,194 (36.5%)	5,772 (34.0%)	5,492 (32.3%)	4,951 (29.1%)	
Current	1,162 (6.8%)	1,526 (9.0%)	1,802 (10.6%)	2,104 (12.4%)	2,832 (16.7%)	
Missing (n)	95	95	87	85	88	
Alcohol status (n%)						<0.001 ^c
Never	631 (3.7%)	676 (4.0%)	655 (3.8%)	755 (4.4%)	974 (5.7%)	
Previous	536 (3.1%)	554 (3.2%)	535 (3.1%)	617 (3.6%)	835 (4.9%)	
Current	15,877 (93.2%)	15,822 (92.8%)	15,847 (93.0%)	15,664 (91.9%)	15,221 (89.4%)	
Missing (n)	41	35	46	45	55	
Physical activity (n%)						<0.001 ^c
Low	2,422 (17.4%)	2,602 (18.8%)	2,653 (19.1%)	2,617 (19.0%)	2,613 (18.9%)	
Moderate	5,879 (42.2%)	5,762 (41.5%)	5,712 (41.1%)	5,533 (40.1%)	5,327 (38.6%)	
High	5,635 (40.4%)	5,506 (39.7%)	5,545 (39.9%)	5,654 (41.0%)	5,854 (42.4%)	
Missing (n)	3,145	3,215	3,175	3,281	3,291	
Fish oil supplementation (Yes%)	48.1	37.7	29.6	22.7	15.5	<0.001 ^c
Missing (n)	56	47	77	82	71	
Comorbidity (Yes%)	38.6	37.3	34.2	33.8	33.0	<0.001 ^c
Plasma omega-3 percentage	6.6 (1.4)	4.9 (0.5)	4.2 (0.4)	3.5 (0.3)	2.6 (0.5)	<0.001 ^b
Plasma omega-6 percentage	36.3 (3.7)	37.4 (3.4)	38.2 (3.3)	39.1 (3.2)	40.2 (3.1)	<0.001 ^b
Plasma DHA percentage	2.8 (0.8)	2.1 (0.5)	1.9 (0.4)	1.7 (0.4)	1.5 (0.4)	<0.001 ^b
Plasma LA percentage	27.0 (3.3)	28.4 (3.1)	29.3 (3.0)	30.2 (3.0)	31.3 (3.1)	<0.001 ^b
Dietary omega-3 percentage	3.6 (1.8)	3.2 (1.4)	3.0 (1.3)	2.9 (1.2)	2.7 (1.0)	<0.001 ^b
Missing (n)	9,355	9,626	9,857	9,974	10,288	
Dietary omega-6 percentage	16.8 (4.6)	16.5 (4.4)	16.6 (4.6)	16.5 (4.7)	16.5 (4.8)	<0.001 ^b
Missing (n)	9,355	9,626	9,857	9,974	10,288	

^a All variables measured at baseline are presented as mean (SD) unless otherwise specified.

^b From the ANOVA test for continuous variables.

^c From the Pearson's Chi-squared test for categorical variables.

Table 1.

Selected participants characteristics at baseline across quintiles of the plasma omega-6/omega-3 PUFAs ratio (n=85,425).

Omega ratio variable forms	Causes of death								
	All-cause			Cancer			Cardiovascular diseases		
	Number of deaths	Partially adjusted associations ^b	Fully adjusted associations ^c	Number of deaths	Partially adjusted associations ^b	Fully adjusted associations ^c	Number of deaths	Partially adjusted associations ^b	Fully adjusted associations ^c
		HR (95% CI)	HR (95% CI)		HR (95% CI)	HR (95% CI)		HR (95% CI)	HR (95% CI)
Continuous	6,461	1.02 (1.02-1.03)	1.02 (1.02-1.03)	2,794	1.02 (1.01-1.03)	1.01 (1.00-1.02)	1,668	1.02 (1.01-1.03)	1.02 (1.01-1.03)
Quintiles (median)									
1 (5.9)	1,348	1.00 (ref)	1.00 (ref)	593	1.00 (ref)	1.00 (ref)	369	1.00 (ref)	1.00 (ref)
2 (7.6)	1,256	1.00 (0.92-1.08)	0.96 (0.88-1.05)	563	1.02 (0.91-1.15)	0.98 (0.86-1.12)	315	0.90 (0.77-1.05)	0.89 (0.75-1.06)
3 (9.1)	1,236	1.06 (0.98-1.15)	1.01 (0.93-1.11)	543	1.08 (0.96-1.21)	0.99 (0.87-1.13)	321	0.97 (0.84-1.13)	0.97 (0.81-1.16)
4 (11.0)	1,252	1.14 (1.06-1.23)	1.09 (1.00-1.19)	548	1.16 (1.03-1.31)	1.12 (0.98-1.27)	306	0.98 (0.84-1.15)	1.02 (0.86-1.22)
5 (14.8)	1,369	1.34 (1.24-1.45)	1.26 (1.15-1.38)	547	1.27 (1.12-1.43)	1.14 (1.00-1.31)	357	1.20 (1.04-1.40)	1.31 (1.10-1.55)
<i>P_{trend}</i>		<0.001	<0.001		<0.001	0.011		0.002	<0.001

Abbreviations: CI, confidence interval; HR, hazards ratio; ref, reference.

^a From Cox proportional hazards regression.

^b Adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre.

^c Adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

Table 2.

Associations^a of the plasma omega-6/omega-3 PUFAs ratio with all-cause, cancer, and CVD mortality risk in the UK Biobank.

cancer and CVD mortality, they also tended to be stronger among current smokers but did not reach statistical significance (P for interaction = 0.13 and 0.12, respectively). The associations with CVD mortality tended to be stronger among participants without comorbidities (P for interaction <0.01). No significant interactions were found for other risk factors (**Figure 2** [↗](#)).

Restricted cubic spline analysis

Restricted cubic spline analysis suggested significant positive associations of the omega-6/omega-3 ratio with all-cause, cancer, and CVD mortality ($P < 0.05$ for all three outcomes, **Figure 3** [↗](#)). Potential nonlinearity in these positive associations was identified for all-cause mortality ($P < 0.05$) and CVD mortality ($P < 0.05$) but not for cancer mortality ($P = 0.12$). The strength of the relationship between the ratio and all-cause mortality appears to remain at a relatively low level before it starts to increase quickly after the ratio exceeds 8. A similar trend with higher uncertainties was observed for CVD mortality.

Omega-3, Omega-6, Joint analysis, and comparison with dietary PUFAs

We further performed analyses to assess whether the associations of the omega-6/omega-3 ratio with mortality outcomes were primarily driven by omega-3 or omega-6 fatty acids. The correlation between omega-3% and omega-6% was relatively low with $r = -0.12$ ($P < 0.01$). Across all models, both the omega-3% and omega-6% were inversely associated with all three mortality outcomes, except for plasma omega-6% with CVD mortality under Model 3 ($P_{\text{trend}} < 0.01$, **Additional file 2: Table S3 and Table S4**). Notably, their associations remained significant when they were included in the same models. On the other hand, the effect sizes of the inverse associations were always bigger for the omega-3% under the fully adjusted Model 3. For example, when comparing those in the highest omega-3% quintile to the lowest quintile, the fully adjusted HRs (95% CI) for all-cause, cancer, and CVD mortality were, respectively, 0.69 (0.63, 0.76), 0.75 (0.65, 0.87), and 0.68 (0.57, 0.82) (**Additional file 2: Table S3**). The corresponding HRs for the omega-6% were 0.77 (0.70, 0.85), 0.80 (0.68, 0.92), and 0.83 (0.68, 1.02) (**Additional file 2: Table S4**). Furthermore, in another joint analysis of the omega-3% and omega-6%, the lowest risk for all-cause and cancer mortality was observed among those in the joint highest categories of the two fatty acids (**Additional file 2: Table S5**). For example, when comparing those in the highest quintiles of the two fatty acids to the group with the joint lowest group, the HRs (95% CI) for all-cause and cancer mortality were, respectively, 0.48 (95% CI, 0.35, 0.67) and 0.53 (95% CI, 0.33, 0.86). In the analysis of dietary PUFAs with mortality, the effect sizes were smaller and less significant compared with those of the corresponding plasma levels. When analyzed as continuous variables, dietary omega-3%, omega-6% and omega-6/omega-3 ratio were all significantly associated with all-cause, cancer, and CVD mortality, in directions consistent with their plasma counterparts. The trend test across the five quintiles revealed significant associations between dietary omega-3% and omega-6/omega-3 ratio with cancer mortality (P for trend <0.001 and =0.002, respectively; **Additional file 2: Table S6**). The correlation between dietary omega-3% and omega-6% was $r = 0.41$ ($P < 0.01$).

Mediation analysis for intermediate biomarkers

The baseline serum biomarkers of the participants across quintiles of the ratio of omega-6/omega-3 were summarized in **Additional file 2: Table S7**. The estimates for the three paths a (association between plasma omega-6/omega-3 ratio and biomarkers), b (association between biomarkers and mortality outcomes), and c (association between plasma omega-6/omega-3 ratio and mortality outcomes), as well as the percentage of indirect effect (a to b) among the overall associations, were reported in **Additional file 2: Table S8 and Figure S1**. These analyses identified cholesterol, LDL, ApoB, and SHBG as potential mediators for the association of plasma omega-6/omega-3 ratio with

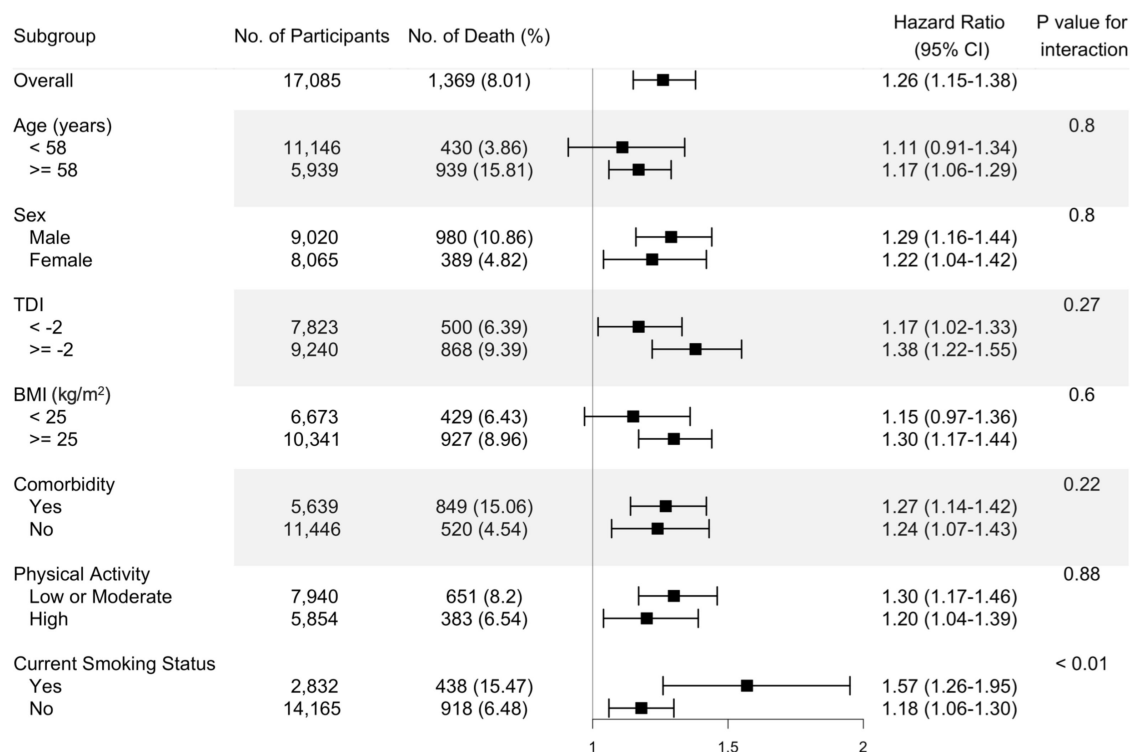


Figure 1.

Risk estimates of all-cause mortality for the highest compared with the lowest quintile of the ratio of plasma omega-6 to omega-3 PUFAs, stratified by potential risk factors. Results were adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

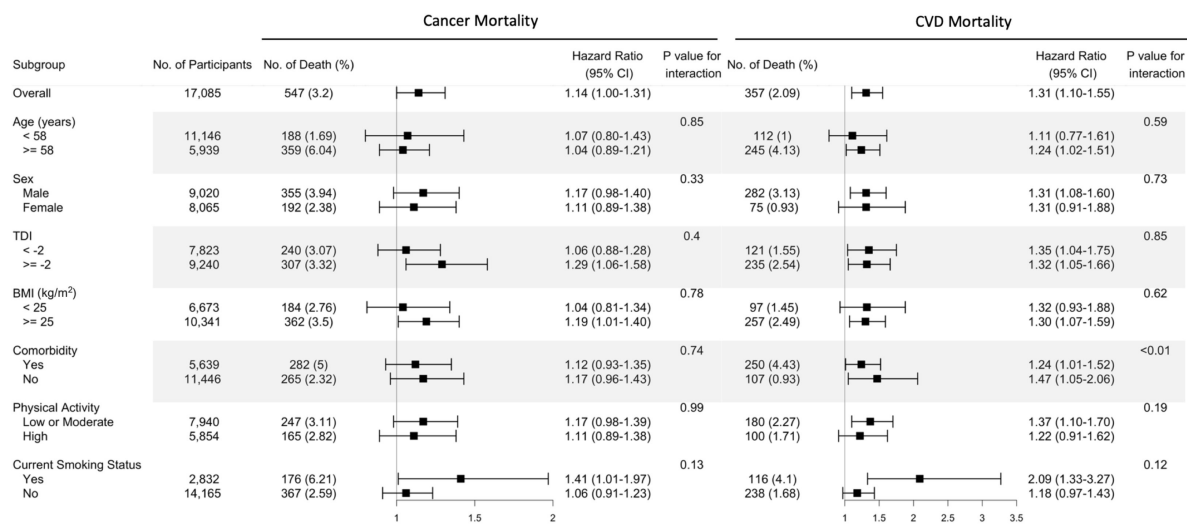


Figure 2.

Risk estimates of cause-specific mortality for the highest compared with the lowest quintile of the ratio of plasma omega-6 to omega-3 PUFAs, stratified by potential risk factors. Results were adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

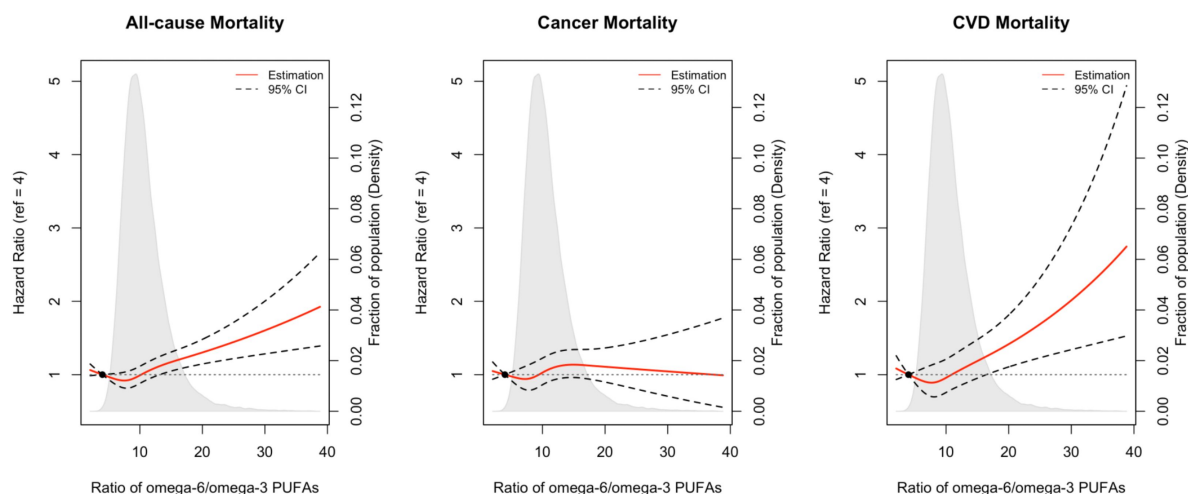


Figure 3.

Associations of the ratio of omega-6/omega-3 PUFAs with all-cause and cause-specific mortality evaluated using restricted cubic splines. Hazard ratios and omega ratios are presented in the vertical and horizontal axis, respectively.

The best estimates and their confidence intervals are presented as solid red lines and dotted black lines, respectively. The ratio 4 was selected as a reference level, and the x-axis depicts the ratio from 0 to 40. Potential nonlinearity was identified for all-cause mortality ($p < 0.05$) and CVD-caused mortality ($p < 0.05$), but not for cancer-caused mortality ($p = 0.12$). All HRs are adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

all-cause mortality (proportion of mediating = 14.7%, 10.8%, 11.0%, and 5.0%, respectively); cholesterol, LDL, ApoA and ApoB as potential mediators for CVD mortality (proportion of mediating = 16.7%, 11.0%, 6.5%, and 9.8%, respectively).

Other sensitivity analyses

We performed analyses to examine the associations of DHA% and LA% with mortalities. The correlation between DHA% and LA% was relatively low with $r = 0.03$ ($P < 0.01$). Under the fully-adjusted model, both DHA% and LA% were inversely associated with all three mortality outcomes, regardless of treated as continuous or quintiles (**Additional file 2: Table S9**). After further adjustment of the fish oil supplementation status, the associations between the ratio of omega-6/omega-3 and mortality outcomes were slightly attenuated yet did not alter the main findings (**Additional file 2: Table S10**). Our primary analysis excluded participants with missing information (i.e., a complete-case analysis). We performed a sensitivity analysis using the multiply-imputed datasets, and there were no substantial changes (**Additional file 2: Table S11**). Moreover, the exclusion of participants who died during their first-year follow-up did not materially alter the results (**Additional file 2: Table S12**). The baseline characteristics are comparable between participants with or without exposure information (**Additional file 2: Table S13**).

Discussion

In this prospective population-based study of UK individuals, we showed that a higher ratio of plasma omega-6/omega-3 fatty acids was positively associated with the risk of all-cause, cancer, and CVD mortality. These associations were independent of most risk factors examined, including age, sex, TDI, BMI, comorbidities, and physical activity, but they were stronger in current smokers. These relationships were linear for cancer mortality but not for all-cause and CVD mortality. For those two outcomes, the risk of mortality first decreased at lower ratios and then increased, with an inflection point around the ratio of 8. Moreover, omega-3 and omega-6 PUFAs in plasma were consistently and inversely associated with all-cause, cancer, and CVD mortality, with omega-3 showing stronger effects.

To date, studies that examined the relationship between the ratio of omega-6/omega-3 PUFAs and mortality in the general population are sparse^{4, 14, 15, 19}. Similar to our finding, in a 2017 report from a prospective women cohort study ($n=6,501$; 1,875 all-cause deaths, 617 CVD deaths, 462 cancer deaths)⁴, the adjusted HR for all-cause mortality was 1.10 (1.02–1.19) per 1-SD increase of omega-6/omega-3 ratio in red blood cells, however, the effects were not significant for cancer and CVD deaths. Another study supporting our results was conducted on elderly Japanese individuals ($n=1,054$; 422 deaths). It found that the ratio of an omega-3 fatty acid, eicosapentaenoic acid, to an omega-6 fatty acid, arachidonic acid (ARA), is inversely associated with all-cause mortality, with an HR of 0.71 (95% 0.53–0.96) comparing the highest to the lowest tertile¹⁴. Findings of previous studies based on dietary intake were null^{15, 19}. A prospective cohort involving 145 hemodialysis patients enrolled in Southern California during 2001–2007 (42 all-cause deaths) showed that the estimated HR (95% CI) for all-cause mortality among those in the lowest relative to the highest quartiles of dietary omega-6/omega-3 ratio was 0.37 (0.14–1.08)¹⁹. Although the effect was not significant, it still shows the protective trend of a lower omega-6/omega-3 ratio against premature death, when taking the sample size and study population into consideration. In a 2019 report based on two population-based prospective cohorts in China ($n=14,117$, 1,007 all-cause deaths) and the US ($n=36,032$, 4,826 all-cause deaths), the effect of the omega-6/omega-3 ratio intake was not significant (HR (95% CI) is 0.95 (0.80–1.14) for China and 0.99 (0.89–1.11) for US, respectively)¹⁵. These discrepancies may be explained by the usage of circulating biomarkers or dietary intakes for calculating the omega-6/omega-3 ratio. Indeed, we performed the same association analyses for dietary PUFAs and found that their associations with

mortality have smaller effect sizes and are less statistically significant, although maintaining the same effect directions as plasma PUFAs. The estimated dietary intakes may be inaccurate due to recall bias or outdated food databases¹⁶. The circulating level is a more objective measurement of PUFA status and thus provides a more reliable picture of the effects of omega-3 and omega-6 PUFAs on mortality. Additionally, we observed that the omega-6/omega-3 ratio tends to have stronger effects on mortality outcomes in current smokers. This is consistent with a previous observation that the association between habitual fish oil supplementation and a lower risk of all-cause mortality is stronger in current smokers²⁶. Moreover, we found that current smokers in UK Biobank tend to have higher omega-6/omega-3 ratios. Similarly, prior studies reported that current smokers have lower circulating omega-3 PUFAs than non-smokers but similar circulating levels of omega-6 PUFAs^{32, 33}. It is likely that omega-3 PUFAs play more important roles in current smokers, resulting in a stronger effect of the omega-6/omega-3 ratio on mortality. However, the underlying mechanism awaits future investigations.

A large number of existing observational studies documented the inverse association of circulating levels and intake of omega-3 PUFAs with mortality^{5, 6, 27, 34–39}, which is in line with our finding that individuals in the highest quintile of the plasma omega-3% had approximately 30% lower risk for all-cause and CVD mortality and 25% lower risk for cancer mortality, when compared to those in the lowest quintile. Our analysis of dietary omega-3% also revealed inverse associations with all-cause and cancer mortality, with 9% and 18% lower risk, respectively. Few previous studies have examined the association in generally healthy populations^{5, 27, 34, 36}. In a 2016 meta-analysis³⁴ of seven epidemiologic studies for dietary omega-3 PUFAs intake and four studies for circulating levels, the estimated relative risk for all-cause mortality was 0.91 (95% CI: 0.84–0.98) when comparing the highest to the lowest categories of omega-3 PUFA intake. In two 2018 reports of population-based studies in the US, the circulating level and dietary intake of omega-3 PUFAs were significantly associated with lower total mortality^{5, 27}. In a 2021 meta-analysis of 17 epidemiologic studies, comparing the highest to the lowest quintiles of circulating EPA and DHA, the estimated HRs for all-cause, CVD, and cancer mortality were 0.84 (0.79–0.89), 0.80 (0.73–0.88) and 0.87 (0.78–0.96), respectively³⁶. Moreover, inverse associations of omega-3 PUFAs biomarkers with total mortality were found in patients with myocardial infarction^{37, 39}, type 2 diabetes¹², and other diseases^{6, 35, 38, 40}; inverse associations of omega-3 PUFAs with CVD mortality were also reported in these patient groups^{6, 12, 39}. However, there are also reports of a null relationship between omega-3 PUFAs and mortality^{13, 14, 41}. One possible explanation for the discrepancies could be the limited statistical power due to the small sample size and the small number of events. Moreover, the inconsistency may be due to unique sample characteristics; one study only involved elder patients¹⁴, and one only involved women⁴¹.

A smaller number of studies have evaluated the associations of omega-6 PUFAs with mortality. Our findings in the large UK prospective cohort are in accordance with the results of several previous studies that increased circulating levels of omega-6 PUFAs were associated with decreased all-cause mortality^{9, 11, 15, 42}. Moreover, a 2019 meta-analysis involving 18 cohort studies and 12 case-control studies showed that an omega-6 PUFA, linoleic acid, was inversely associated with CVD mortality; the HR per interquintile range was 0.78 (95% CI, 0.70–0.85)¹⁰. Other studies, however, did not support such inverse associations^{4, 12–14, 41}. Although the findings were inconsistent, no previous studies reported harmful effects of omega-6 PUFAs on mortality^{4, 11–15, 41–44}. Our studies support the protective effects of omega-6 PUFAs on all-cause and cause-specific mortality. Further research is needed to investigate the health impacts of omega-6 PUFAs in laboratory studies, epidemiological investigations, and clinical trials.

We observed that the omega-6/omega-3 ratio is positively, while both of its numerator and denominator, omega-6% and omega-3%, are negatively associated with mortality. Our findings support that both omega-6 and omega-3 PUFAs are protective against death and that the positive associations of the omega-6/omega-3 ratio with mortality outcomes are likely due to the stronger

effects of omega-3 than omega-6 PUFAs. Our mediation analysis with clinical biomarkers for CVD and cancer revealed that cholesterol, LDL, and ApoB partially mediate the effects of the omega-6/omega-3 ratio on all-cause and CVD mortality. However, the proportions mediated are relatively small (~10%), suggesting the presence of direct effects or unexamined factors. Our findings call for future studies that evaluate the utility and investigate the mechanisms of the omega-6/omega-3 ratio.

Strengths of our current study include the use of objective PUFA biomarkers in plasma instead of the estimated intakes from dietary questionnaires, which increases the accuracy of exposure assessment. Moreover, the prospective population-based study design, large sample size, long duration of follow-up, and detailed information on potential confounding variables, substantially mitigate the possible complications from reverse causality and confounding bias. In several sensitivity analyses, most of the documented associations remain materially unchanged, indicating the robustness of our results.

Several potential limitations deserve attention. First, plasma omega-3 and omega-6 PUFAs were measured only once at baseline. Their levels may vary with diet or other lifestyle factors, which could cause misclassification over follow-up. However, some studies demonstrated that multiple measurements of omega-3 PUFAs have been consistent for a 6-month period⁴⁵. Moreover, the 13-year within-person correlation for circulating omega-6 PUFAs was comparable to such correlations for other major CVD risk factors⁴⁶. Thus, the single measurement of PUFAs at baseline, although not perfect, provides us with adequate information to investigate the relative long-term effects. Second, although we adjusted for many potential confounders in the model, we cannot rule out the imprecisely measured and unmeasured factors. Last, although we included individuals of different ancestries in the analysis, over 90% of the participants were of European ancestry. The generalizability of our findings across ancestries requires future verification.

Conclusions

In this large prospective cohort study, we documented robust positive associations of the plasma omega-6/omega-3 fatty acids ratio with the risk of all-cause, cancer, and CVD mortality. Moreover, we found that plasma omega-3 and omega-6 PUFAs were independently and inversely associated with the three mortality outcomes, with omega-3 fatty acids showing stronger protective effects. Our findings support the active management of a high circulating level of omega-3 fatty acids and a low omega-6/omega-3 ratio to prevent premature death.

Declarations

Ethics approval and consent to participate

The UK Biobank received ethical approval from the research ethics committee (reference ID: 11/NW/0382). Written informed consent was obtained from participants.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analyzed during the current study are available from the UK Biobank through an application process (www.ukbiobank.ac.uk/).

Competing interests

The authors declare no competing interests.

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Authors' contributions

YZ performed data analysis, prepared visualizations, and wrote the original draft of the manuscript. YS, YQ, and SS contributed to the data analysis. YS (yeshen@uga.edu) and KY (kaixiong.ye@uga.edu) contributed equally to this project and should be considered co-corresponding authors. They jointly designed and supervised the project. JTB contributed to the interpretations of results. All authors critically edited the manuscript for important intellectual content. The corresponding author (KY) attests that all listed authors meet the authorship criteria and that no others meeting the criteria were omitted.

Data Availability

The datasets analyzed during the current study are available from the UK Biobank through an application process (www.ukbiobank.ac.uk/ .

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Abbreviations

- 95% CI
95% confidence interval
- ANOVA
Analysis of variance
- ApoA
Apolipoprotein A
- ApoB
Apolipoprotein B
- BMI
Body mass index
- CRP
C-creative protein

- Cardiovascular disease
- DHA
Docosahexaenoic acid
- DHA%
Docosahexaenoic acid to total fatty acids percentage
- E2
Oestradiol
- EPA
Eicosapentaenoic acid
- HDL-C
High density lipoprotein-cholesterol
- HR
Hazard ratio
- LA
Linoleic acid
- LA%
Linoleic acid to total fatty acids percentage
- IGF-1
Insulin-like growth factor 1
- LDL
Low density lipoprotein
- Lp(a)
Lipoprotein A
- Omega-3%
Omega-3 fatty acids to total fatty acids percentage
- Omega-6%
Omega-6 fatty acids to total fatty acids percentage
- PUFAs
Polyunsaturated fatty acids
- Ref
Reference
- SD
Standard deviation
- SHBG
Sex hormone binding globulin
- TG
Triglyceride
- TTST
Testosterone

Supplementary Information

Additional file 1: Table S1. Literature review.

Additional file 2: Table S2. Risk estimates of ratio of plasma omega-6 to omega-3 PUFAs with all-cause, cancer and CVD mortality, stratified by potential risk factors, in the UK Biobank Study.

Table S3. Associations of plasma omega-3 PUFAs percentage with all-cause, cancer, and CVD mortality risk in the UK Biobank Study. **Table S4.** Associations of plasma omega-6 PUFAs percentage with all-cause, cancer, and CVD mortality risk in the UK Biobank Study. **Table S5.** Fully adjusted joint associations of plasma omega-3 PUFAs percentage and omega-6 PUFAs percentage with all-cause and cause-specific mortality in the UK Biobank Study. **Table S6.** Associations of the dietary PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank. **Table S7.** Selected participants serum biochemical markers at baseline across quintiles of the plasma omega-

6/omega-3 PUFAs ratio. **Table S8.** Mediation analysis of biomarkers on associations between plasma omega-6/omega-3 ratio and mortality. **Figure S1.** Directed acyclic graph to explain mediation. **Table S9.** Associations of DHA and LA with all-cause, cancer, and CVD mortality risk in the UK Biobank. **Table S10.** Associations of ratio of omega-6/omega-3 PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank, covariates including fish oil supplementation status. **Table S11.** Associations of ratio of omega-6/omega-3 PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank Study, with multiple imputation for missing data. **Table S12.** Associations of ratio of omega-6/omega-3 PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank Study, excluding those who died in the first follow-up year. **Table S13.** Baseline characteristics of participants with missing exposure information and not included in the study.

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Editors

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Reviewer #1 (Public Review):

Summary: The manuscript offers a commendable exploration into the relationship between plasma omega-6/omega-3 fatty acid ratios and mortality outcomes.

Strengths: The chosen study design and analytical techniques align well with the research objectives, and the results resonate with existing literature.

Weaknesses: Lack of information on the selection criteria for participants; 5. The analysis of individual PUFAs is not appropriate; The definition of comorbidities is vague; The rationale of conducting the mediation analysis of blood biomarkers is not given.

Reviewer #2 (Public Review):

Summary: This study utilized a large sample from the UK Biobank which enhanced statistical robustness, employed a prospective design to establish clear temporal relationships, used objective biomarkers for assessing plasma omega-6/omega-3 ratio, and investigated various mortality causes including CVD and cancer for a holistic health understanding.

Strengths: The authors used a large sample size, employed a prospective design, and investigated various mortality.

Weaknesses: Analyzing n-3 and n-6 PUFAs separately might be less instructive. It might not be methodologically sound to treat TG, HDL, LDL, and apolipoproteins as mediators. It's imperative to exercise caution when drawing causal conclusions from the observed correlations. The manuscript might propose potential research trajectories.

Reviewer #3 (Public Review):

Summary: The authors are trying to find out whether the levels of omega-6 and omega-3 fatty acids in the blood are linked to the likelihood of dying from anything, of dying from cancer and of dying from cardiovascular disease. They use a large dataset called UK Biobank where fatty acid levels were measured in blood at the start of the study and what happened to the participants over the following years (average of 12.7 years) was followed. They find that both omega-6 AND omega-3 fatty acids were linked with less likelihood of dying from anything, from cancer and from cardiovascular disease. The effects of omega-3s were stronger. They then made a ratio of omega-6 to omega-3 fatty acids and found that as that ratio increased risk of dying also increased,. This supports the idea that omega-3s have stronger effects than omega-6s.

Strengths: This is a large study (over 85,000 participants) with a good follow up period (average 12.7 years). Using blood levels of fatty acids is superior to using estimated dietary intakes. The authors take account of many variables that could interfere with the findings (confounding variables) - they do this using statistical methods.

Weaknesses: There are several omega-6 and omega-3 fatty acids - it is not clear which ones were actually measured in this study.